

Week 2, September 9, 2026

Room The course will from week 2 and on take place in 303a/45.

Main topics

- Numerical implementation of model problems
- Singular value decomposition of operators and matrices
- Truncated SVD
- Why are inverse problems challenging?

Material

From last week: Section 1.1-1.2 in [Kirsch] and section 3.1 in [PCH].

New material: Kirsch 2-2.1 (you may skip 27-35). Appendix A, in particular Definition A.56, Theorem A.57, Theorem A.58. You can think of compact operators as integral operators.

Program

We will go with the tentative program:

13:00-13:30 Ill-posedness and model problems

13:40-15:40 Exercise session - discretization of model problems

15:50-17:00 SVD for matrices and compact operators; regularization

Exercises

You are highly encouraged to discuss and solve the exercises in small groups.

In the exercise session today we will implement the model problems and approximate the forward operator K by a matrix A .

1. Warm up: (spend less than 15 minutes)

The numerical differentiation problem is an inverse problem (see slides from the morning in week 1). The problem is implemented in the shared notebook

`NumericalDifferentiation.ipynb` to be found in learn. Try out the code.

- Change the function to $y(s) = s^2 e^{-s}$.
- Solve the inverse problem and compare to the true solution.
- Experiment with the noise level and the number of discretization points.

2. For the Abel problem: Show that for $\nu < -1/2$, $x(s) = s^\nu$ produce $Kx(t) = y(t) = B(1 + \nu, 1/2)t^{\nu+1/2}$, with $B(z_1, z_2)$ denoting the beta function.
3. Abel operator by quadrature and collocation (Nyström method):
The operator

$$Kx(t) = \frac{1}{\sqrt{\pi}} \int_0^t \frac{x(s)}{\sqrt{t-s}} ds$$

$$= \int_0^1 k(t, s)x(s) ds \quad \text{with } k(t, s) = \begin{cases} \frac{1}{\sqrt{\pi(t-s)}}, & s < t \\ 0, & s > t \end{cases}$$

is approximated by a matrix A as follows: Take $h = 1/N$ and define

s -grid: $s_1 = h/2, s_2 = h/2 + h, \dots, s_N = h/2 + (N-1)h$

t -grid $t_1 = h, t_2 = 2h, \dots, t_N = Nh = 1$.

Then set $A_{ij} = hk(t_i, s_j)$, i.e. apply

$$Kf(t_i) \approx h \sum_j k(t_i, s_j)f(s_j).$$

- Implement the method numerically to get the matrix A . Experiment with different functions x .
 - Logplot the singular values of the A .
 - Try having similar t and s grids. Find a way for handling the singularity.
4. Heat equation by discrete Fourier transform
We will use the variables (s, t) ; s is the spatial variable for initial condition $x(s)$, t is the spatial variable for the terminal condition $y(t)$. Recall the formula

$$Kx(t) = y(t) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} c_n e^{-Tn^2} \sin(nt),$$

with

$$x(s) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} c_n \sin(ns)$$

being the initial condition with Fourier (sine) coefficients c_n .

Put differently, the operator K is

$$y(t) = Kx(t) = \int_0^\pi k(t, s)x(s) ds$$

with

$$k(t, s) = \frac{2}{\pi} \sum_{n=1}^{\infty} e^{-Tn^2} \sin(nt) \sin(ns).$$

- Set up a complete code that given initial function x approximates y (both are sampled and represented as vectors) using a Galerkin method with the ONB $\phi_j(t) = \psi_j(t) = \sin(nt)$. Use FFT to obtain the coefficients.
- Experiment with different choices of initial conditions x and compute an approximation of y . Look at different choices of T .
- The problem is linear and can be represented by a matrix A . Compute the matrix.
- Logplot the singular values of A .
- Repeat the above for different levels of discretization.